

Program:

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#include <stdio.h>
#include <stdlib.h>
// Node structure definition
struct Node {
    int data;
    struct Node* next;
};
// Function prototypes
struct Node* createNode(int data);
void push(struct Node** top, int data);
int pop(struct Node** top);
void displayStack(struct Node* top);
int main() {
    struct Node* top = NULL;
    int choice, element;
    while (1) {
        printf("\nStack Operations Menu:\n");
        printf("1. Push\n");
        printf("2. Pop\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter element to push: ");
                scanf("%d", &element);
                push(&top, element);
                break;
            case 2:
                element = pop(&top);
                if (element != -1)
                    printf("Popped element: %d\n", element);
                break;
            case 3:
                displayStack(top);
                break;
            case 4:
                exit(0);
            default:
                printf("Invalid choice! Please enter a valid option.\n");
        }
    }
}
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}
return 0;
}
// Function to create a new node
struct Node* createNode(int data) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    if (!newNode) {
        printf("Memory allocation error\n");
        exit(1);
    }
    newNode->data = data;
    newNode->next = NULL;
    return newNode;
}
// Function to push an element onto the stack
void push(struct Node** top, int data) {
    struct Node* newNode = createNode(data);
    newNode->next = *top;
    *top = newNode;
    printf("Element pushed: %d\n", data);
}
// Function to pop an element from the stack
int pop(struct Node** top) {
    if (*top == NULL) {
        printf("Error: Stack underflow. Cannot pop element.\n");
        return -1;
    }
    struct Node* temp = *top;
    int poppedElement = temp->data;
    *top = (*top)->next;
    free(temp);
    return poppedElement;
}
// Function to display the stack elements
void displayStack(struct Node* top) {
    if (top == NULL) {
        printf("Stack is empty.\n");
        return;
    }
    printf("Stack elements: ");
    struct Node* temp = top;
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
}

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}  
printf("\n");  
}
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